POWER TREE

J1 · Pogo
magnetic 4-pin
VBUS
GND

VBUS_5V

U10 · BQ25180
1S linear charger + power path
IN
SYS
BAT
ISET/ILIM per 370 mAh (0.5C)

VSYS

U11 · TPS63802
buck-boost → 3.3 V
VIN
VOUT
1 µH · 2×22 µF out

+3V3

loads: U1 · U3 · U4 · U5 · U6 VBAT · U8 · U9 VCI/VDDIO · U13 · U14

VBAT → U13 MAX17048 CELL

BT1 · LiPo 370 mAh
w/ PCM · 3.0–4.2 V

U12 · TLV75518P
LDO 1.8 V

+1V8

→ U6 VDD(PAD), U7 VREF1

OPTIONAL · AMOLED ELVDD/ELVSS
only if the panel FPC doesn't generate them
check panel datasheet

SIGNAL INTERCONNECT

U1 · STM32U5A9VJT6Q
LQFP-100 · Cortex-M33 160 MHz · 2.5 MB SRAM

U3 · MAX-M10S
GNSS · 3V3
TXD — GNSS_TX — USART2_RX
RXD — GNSS_RX — USART2_TX
TIMEPULSE — GNSS_PPS — TIM2_CH1
RESET_N — GNSS_RST_N — GPIO

U4 · SX1262 module
E22-900M22S · LoRa 915 · 3V3
SCK — LORA_SCK — SPI1_SCK
MOSI — LORA_MOSI — SPI1_MOSI
MISO — LORA_MISO — SPI1_MISO
NSS — LORA_NSS — GPIO
BUSY — LORA_BUSY — GPIO in
DIO1 — LORA_DIO1 — EXTI
NRESET — LORA_NRST — GPIO
RXEN — LORA_RXEN — GPIO
TXEN — LORA_TXEN — GPIO

U5 · DWM3000
UWB ch5 · 3V3 · Stretch 2
SPICLK — UWB_SCK — SPI2_SCK
SPIMOSI — UWB_MOSI — SPI2_MOSI
SPIMISO — UWB_MISO — SPI2_MISO
SPICSn — UWB_CS — GPIO
IRQ — UWB_IRQ — EXTI
WAKEUP — UWB_WAKE — GPIO
RSTn — UWB_RSTN — GPIO OD

U6 · PN7150
NFC · VBAT 3V3 · pads 1V8
SCL — NFC_SCL — I2C1_SCL
SDA — NFC_SDA — I2C1_SDA
IRQ — NFC_IRQ — EXTI
VEN — NFC_VEN — GPIO
via U7 PCA9306 3V3↔1V8
I2C addr 0x28

Power pins
VDD ×n, VDDA, VDDUSB → +3V3
100 nF per VDD pin + 4.7 µF bulk
VDDSMPS → +3V3, VLXSMPS → 2.2 µH
VDD11 ×2 → 2.2 µF each

Clocks
PC14/PC15 → Y1 32.768 kHz LSE
PH0/PH1 → Y2 16 MHz HSE

Boot / reset
BOOT0 → 10 kΩ to GND
NRST → 100 nF + J2 pad

Interrupt inputs (▶): 9
each on a different pin number;
EXTI lines are shared across ports

U8 · BNO085
9-DOF fusion IMU · SPI mode
SPI3_SCK — IMU_SCK — H_SCL/SCK
SPI3_MOSI — IMU_MOSI — SA0/MOSI
SPI3_MISO — IMU_MISO — H_SDA/MISO
GPIO — IMU_CS — H_CSN
EXTI — IMU_INT — H_INTN
GPIO — IMU_RST — NRST
GPIO — IMU_WAKE — PS0/WAKE
PS1 = 1 (strap), BOOTN 10k↑ + pad

U9 · 1.43" AMOLED
466×466 · CO5300 · CTP
OCTOSPI1_CLK — DISP_CLK — QSPI_CLK
OCTOSPI1_NCS — DISP_CS — QSPI_CS
OCTOSPI1_IO0 — DISP_D0 — QSPI_D0
OCTOSPI1_IO1 — DISP_D1 — QSPI_D1
OCTOSPI1_IO2 — DISP_D2 — QSPI_D2
OCTOSPI1_IO3 — DISP_D3 — QSPI_D3
EXTI — DISP_TE — TE
GPIO — DISP_RST — RESET
EXTI — TP_INT — TP_INT
GPIO — TP_RST — TP_RST
touch SDA/SCL on I2C2

I2C2 shared bus
4.7 kΩ pull-ups to 3V3 · 400 kHz
I2C2_SCL — I2C2_SCL — SCL
I2C2_SDA — I2C2_SDA — SDA
GPIO — HAPT_EN — U14 EN
EXTI — FG_ALRT — U13 ALRT
EXTI — CHG_INT — U10 INT
U14 DRV2605L 0x5A → LRA M1
U13 MAX17048 0x36
U10 BQ25180 0x6A
U9 touch ctrl (addr per panel)

SW1 · PEC06 crown
6 ppr encoder + push
TIM3_CH1 — ENC_A — A
TIM3_CH2 — ENC_B — B
EXTI — ENC_SW — SW
10k↑ + 10 nF per Bourns filter

J1 · Pogo dock
4-pin magnetic
USART1_TX — DOCK_TX — TX
USART1_RX — DOCK_RX — RX
VBUS/GND → U10 IN
TVS on all 4 pins

J2 · TC2030 SWD
Tag-Connect footprint
SWDIO — SWDIO — SWDIO
SWCLK — SWCLK — SWCLK
+ NRST, 3V3, GND pads

LEGEND

- signal net (label = net name)
- power rail
- ▶ interrupt input at U1 (EXTI)
- ○ open-drain output (never drive high)
- optional / panel-dependent

BandLink · NFC GPS Bracelet
System schematic · interconnect level

Scope: MVP + Stretch 1 + Stretch 2 · 53 signal pins	
Sheet 1 / 1	Rev 2 · 2026-10-22
CECS 490A · Team S02-T01	Pin numbers: assign in CubeMX

Net list

NET	U1 FUNCTION	TO	TYPE	NOTES
GNSS_TX	USART2_RX	U3 TXD	UART	
GNSS_RX	USART2_TX	U3 RXD	UART	
GNSS_PPS	TIM2_CH1	U3 TIMEPULSE	Timer	Timer input capture, disciplines TDMA slot clock
GNSS_RST_N	GPIO	U3 RESET_N	GPIO	Also bring SAFEBOOT_N to a test pad
LORA_SCK	SPI1_SCK	U4 SCK	SPI/QSPI	
LORA_MOSI	SPI1_MOSI	U4 MOSI	SPI/QSPI	
LORA_MISO	SPI1_MISO	U4 MISO	SPI/QSPI	
LORA_NSS	GPIO	U4 NSS	GPIO	
LORA_BUSY	GPIO in	U4 BUSY	GPIO	Poll low before every SPI command
LORA_DIO1	EXTI	U4 DIO1	EXTI in	TxDone / RxDone / Timeout IRQ
LORA_NRST	GPIO	U4 NRESET	GPIO	
LORA_RXEN	GPIO	U4 RXEN	GPIO	Drop TXEN if module lets DIO2 drive the RF switch
LORA_TXEN	GPIO	U4 TXEN	GPIO	
UWB_SCK	SPI2_SCK	U5 SPICLK	SPI/QSPI	
UWB_MOSI	SPI2_MOSI	U5 SPIMOSI	SPI/QSPI	
UWB_MISO	SPI2_MISO	U5 SPIMISO	SPI/QSPI	
UWB_CS	GPIO	U5 SPICSn	GPIO	
UWB_IRQ	EXTI	U5 IRQ	EXTI in	
UWB_WAKE	GPIO	U5 WAKEUP	GPIO	Wakes DW3000 from deep sleep
UWB_RSTN	GPIO OD	U5 RSTn	GPIO OD	Open-drain only; DW3000 RSTn must not be driven high
NFC_SCL	I2C1_SCL	U6 SCL	I2C	Own bus, through U7 level shifter (pads 1.8 V)
NFC_SDA	I2C1_SDA	U6 SDA	I2C	Own bus, through U7 level shifter (pads 1.8 V)
NFC_IRQ	EXTI	U6 IRQ	EXTI in	Level-shift 1.8 V → 3.3 V input
NFC_VEN	GPIO	U6 VEN	GPIO	Low = hard power-down between pairings
IMU_SCK	SPI3_SCK	U8 H_SCL/SCK	SPI/QSPI	
IMU_MOSI	SPI3_MOSI	U8 SA0/MOSI	SPI/QSPI	
IMU_MISO	SPI3_MISO	U8 H_SDA/MISO	SPI/QSPI	
IMU_CS	GPIO	U8 H_CSN	GPIO	
IMU_INT	EXTI	U8 H_INTN	EXTI in	Data-ready / significant-motion wake
IMU_RST	GPIO	U8 NRST	GPIO	
IMU_WAKE	GPIO	U8 PS0/WAKE	GPIO	PS0 doubles as WAKE in SPI mode
DISP_CLK	OCTOSPI1_CLK	U9 QSPI_CLK	SPI/QSPI	
DISP_CS	OCTOSPI1_NCS	U9 QSPI_CS	SPI/QSPI	
DISP_D0	OCTOSPI1_IO0	U9 QSPI_D0	SPI/QSPI	
DISP_D1	OCTOSPI1_IO1	U9 QSPI_D1	SPI/QSPI	
DISP_D2	OCTOSPI1_IO2	U9 QSPI_D2	SPI/QSPI	
DISP_D3	OCTOSPI1_IO3	U9 QSPI_D3	SPI/QSPI	
DISP_TE	EXTI	U9 TE	EXTI in	Start frame DMA on TE to avoid tearing
DISP_RST	GPIO	U9 RESET	GPIO	
TP_INT	EXTI	U9 TP_INT	EXTI in	
TP_RST	GPIO	U9 TP_RST	GPIO	
I2C2_SCL	I2C2_SCL	SCL	I2C	4.7 kΩ to 3V3, shared by U9 touch, U10, U13, U14
I2C2_SDA	I2C2_SDA	SDA	I2C	4.7 kΩ to 3V3
HAPT_EN	GPIO	U14 EN	GPIO	
FG_ALRT	EXTI	U13 ALRT	EXTI in	Low-SOC alert, open-drain, 10 kΩ pull-up
CHG_INT	EXTI	U10 INT	EXTI in	Charge status / fault, open-drain, 10 kΩ pull-up
ENC_A	TIM3_CH1	SW1 A	Timer	Timer encoder mode, no CPU load while counting
ENC_B	TIM3_CH2	SW1 B	Timer	
ENC_SW	EXTI	SW1 SW	EXTI in	Long-press = enter NFC pairing
DOCK_TX	USART1_TX	J1 TX	UART	USART1 so the dock can reach the ST system bootloader (check AN2606 pins)
DOCK_RX	USART1_RX	J1 RX	UART	
SWDIO	SWDIO	J2 SWDIO	GPIO	
SWCLK	SWCLK	J2 SWCLK	GPIO	

Part notes for schematic capture

U3 MAX-M10S

- RF_IN: 50 Ω trace to the L1 chip antenna; integrated LNA + SAW, no external LNA.
- V_BCKP: feed from a small supercap or keep-alive rail for hot starts.
- SAFEBOOT_N and EXTINT to test pads.

U4 SX1262 module

- Supply from +3V3 only. SX1262 VBAT max is below a full cell.
- ANT pad: 50 Ω to the 915 MHz antenna, with a u.FL test footprint.
- If the module routes DIO2 to its RF switch, drop LORA_TXEN.

U5 DWM3000

- Respect the module's antenna keep-out at the board edge.
- RSTn is open-drain from U1; never drive it high.
- Decouple close to VDD per the module datasheet.

U6 PN7150 + U7 PCA9306

- VBAT from +3V3, VDD(PAD) from +1V8. Confirm the pad-voltage variant before layout.
- U7 shifts NFC_SCL/SDA; NFC_IRQ needs a 1.8 → 3.3 V buffer.
- Antenna: EMC filter + matching caps with tuning pads; ferrite sheet between coil and battery.

U8 BNO085

- PS1 strapped high; PS0/WAKE has a 10 kΩ pull-up so it reads high at reset (SPI mode).
- BOOTN: 10 kΩ pull-up plus test pad for firmware updates.
- 32.768 kHz crystal on XIN32/XOUT32. Keep away from the LRA and battery.

U9 AMOLED

- Keep QSPI traces short and length-matched.
- Check whether the FPC needs external ELVDD/ELVSS (dashed block).
- Touch controller rides I2C2; TP_INT and TP_RST to U1.

U10 BQ25180 · U11 TPS63802 · U12 TLV75518P

- Charge current about 185 mA (0.5C) for the 370 mAh cell.
- TS pin: 10 kΩ NTC on the cell, or bias it per datasheet if you skip temperature sensing.
- TPS63802: follow the datasheet layout; tight loop on the inductor and output caps.

U13 MAX17048 · U14 DRV2605L · SW1 · J1

- MAX17048 CELL ties straight to the battery terminal, not VSYS.
- DRV2605L in LRA mode, auto-calibrate per motor; IN/TRIG to GND if unused.
- PEC06: 10 kΩ pull-ups and 10 nF caps per the Bourns filter circuit.
- J1: TVS diode on every pogo pin.

Open before layout

- Run the pin assignment in STM32CubeMX: confirm OCTOSPI quad, SPI1–3, I2C1–2, USART1–2, TIM2/TIM3 all route in LQFP-100, with the 9 interrupt inputs on distinct pin numbers.
- Get the exact AMOLED panel datasheet: driver IC, whether ELVDD/ELVSS is on the FPC, connector pinout.
- Confirm the PN7150 order code's pad voltage. If it's 3.3 V tolerant, drop U7 and U12.
- Confirm the battery's pulse discharge rating against the 453 mA peak.